Regular halving, cubic toroids and the limits of preservation

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Abstract

We give a corrected account of the principal results in the inspected preprint on generalized halving by Piedade and Tranchida. The literal coset identification holds without residual connectivity of the output. The latter property instead has an exact parity-shadow criterion, and a finite affine regular hypertope shows that this criterion cannot be omitted. For cubic toroids we determine the actual period lattices of the halving groups, compare all displayed presentations, and prove every allowable geometric branch and diagram. The source's constructions and retained classical inputs are distinguished from the corrective proofs.

1 The complete corrected contract

The source throughout is arXiv:2405.19050v1, dated 29 May 2024 [1]. Its 43 named items and the unnumbered classification in Figures 5–8 are routed individually in the accompanying source map. Repeated introductory statements are not additional results. The complete Topic C common paper [3] and its supplement [4] are both dependencies of this application. The former is retained in full, including its binary-module arguments; its cost cannot be replaced by only the particular theorems invoked here.

The following distinctions determine the mathematical contract. Construction 3.6 is always a geometry and preserves thinness and the natural chamber-transitive action under its stated hypotheses. Residual connectivity requires (PC), and full residue transport then holds. Thus the unrestricted residual-connectivity clauses in Theorem A, Proposition 3.13 and Theorem 3.18 are false. The complete regular counterexample below proves that regular thin input does not remove this obstruction. Theorem B and the literal assertion of Theorem 4.4, which identify the geometric construction with its halving-group coset system, remain true by the maximal-stabilizer proof in the supplement. A coset identification alone does not assert marked subgroup IP. The introductory flag-transitivity converse in Theorem A is also false: the final section gives a connected nonbipartite rank-three example whose halving is flag-transitive although its original geometry is not. This example satisfies (PC), so it is a separate obstruction, and it does not affect the forward action statement in Proposition 3.14.

The general criterion for retaining another leaf is corrected from "both endpoint shadows" to "at least one endpoint shadow" over a bipartite edge shadow. The source criterion is valid for a flag-transitive input, since the endpoints then have equal shadow bipartiteness. The supplement gives the complete five-cycle example separating these assertions. Source Lemma 4.6 has an impossible literal antecedent in the regular setting, rather than a counterexample to its logical implication; the exact induced-neighbor coset criterion replaces it. Source Examples 4.8 and 4.9 are proved there with their hypotheses kept distinct.

For cubic toroids the classical input is the faithful cubical realization and classification retained in source Theorem 5.1 from McMullen–Schulte [2]. The affine calculation below proves exact marked groups and complete corrected presentations for every listed parameter. Certain single extra relations in Propositions 5.8 and 5.11–5.13 present groups larger than intended. The weaker assertion that the intended group is a quotient satisfying those relations remains true. The final geometric proof supplies every nondegeneracy, parity-compatibility and exhaustion argument for Figures 5–8, independently of those insufficient presentations.

The borrowed bipartite theorem, source Theorem 4.1, is used only through its geometric, faithful-action and regular conclusions proved in the common papers. Its arbitrary nonthin full-automorphism semidirect-product assertion is a retained borrowed boundary, not a newly recovered theorem or a premise of these arguments. In the regular case the original endpoint involution a supplies the complement to the index-two

color subgroup, so the splitting used here is explicit. The informal remark that the hemicube phenomenon seems sporadic is left as an observation; no classification or open-question solution is claimed. The later journal body is outside the inspected full-text boundary.

The arguments have the following provenance. The affine counterexample is the accepted phase30 proof, reproduced completely below with its finite certificate. The period-intersection and geometric-branch proofs are the separately reviewed N014 packets, reproduced here with only section and label changes. The common supplement similarly reuses the phase27, stabilizer and further-leaf proofs. This assembly supplies an organized dependency and source contract, not a new claim of priority or a new authorship claim for inherited arguments.

2 A finite affine construction and its hypotheses

Let M be a finite, thin, residually connected rank-three incidence geometry with types called vertices, edges and faces. Let K act regularly on its chambers. Assume the following concrete map conditions: every edge has two distinct incident vertices and two distinct incident faces; every incident vertex-face pair has exactly two common edges; and an edge common to a vertex and a face makes that vertex-face pair incident. These conditions hold for the seed below and are checked directly, rather than inferred merely from subgroup IP.

Write Ω for the edge set. For a seed vertex v let B_v be its incident-edge set, and for a seed face f let D_f be its incident-edge set. Let C be a K-invariant binary linear code in $\mathbb{F}_2^{\Omega}$, and put $N = \mathbb{F}_2^{\Omega}/C$. Write q_e for the image of the coordinate vector at e. For any block B among the B_v and D_f put $W_B = \mathrm{span}\{q_e : e \in B\}$. Require:

1. All q_e are nonzero and distinct.
2. For every block B, $q_e \in W_B$ if and only if $e \in B$.
3. For every incident seed vertex-face pair (v,f), $W_{B_v} \cap W_{D_f} = \mathrm{span}\{q_e : e \in B_v \cap D_f\}$.

The quotient columns span N by construction. No assumption about intersections for nonincident seed vertex-face pairs is needed.

Define a typed incidence system Γ as follows. Its type-0 objects are vectors $n \in N$. Its type-1 objects are pairs $(e, n + \langle q_e \rangle)$, identified with the graph edge $\{n, n + q_e\}$. Its type-2 objects are pairs $(v, n + W_{B_v})$; its type-3 objects are pairs $(f, n + W_{D_f})$. The seed labels v and f remain part of each object, even if two affine subspaces happen to coincide.

Vertex-edge incidence is endpoint containment. A vertex is incident with a higher object exactly when it belongs to its affine subspace. An edge (e,L) is incident with (v,S), respectively (f,T), exactly when $e \in B_v$ and $L \subseteq S$, respectively $e \in D_f$ and $L \subseteq T$. Finally, (v,S) and (f,T) are incident exactly when **v and f are incident in the seed and** $S \cap T \neq \varnothing$. Distinct objects of the same type are not incident.

2.1 Affine construction theorem

Under these hypotheses Γ is a finite regular hypertope, with regular group $G = N \rtimes K$. It satisfies (B1) and (B2) at types {0,1}.

Here regular hypertope means a thin, residually connected geometry with a type-preserving group acting regularly on chambers. In fact G is its full type-preserving automorphism group.

Geometry and chambers. A chamber is uniquely specified by a vector n and a seed chamber (v,e,f): its four objects are n, $(e, n + \langle q_e \rangle)$, $(v, n + W_{B_v})$, and $(f, n + W_{D_f})$. Every flag extends to such a chamber. For a flag containing a vertex, use its vector and extend the prescribed seed flag in M. For a flag containing an edge but no vertex, choose either endpoint; all prescribed higher affine subspaces contain this endpoint, and its prescribed seed data form a flag. A flag containing only an incident higher pair has a common vector by definition and two possible common seed edges. A single higher object contains a vector and has an incident seed chamber. The empty flag also extends. This proves the geometry axiom, with no coset-Helly assumption.

Thinness. A panel missing type 0 has the two endpoints of its edge as completions. A panel missing type 1 has its fixed vector and incident seed vertex-face pair, hence exactly their two common edge labels. They give distinct edges by hypothesis 1. A panel missing type 2 has the two possible seed vertices at its prescribed edge-face pair; each determines exactly one affine coset through the fixed vector. The dual argument treats type 3. Thus every panel has two completions.

Residual connectivity. The residue at a vector n is exactly M, with types reordered as edges, vertices, faces: each seed higher label has a unique affine coset through n. Thus all flags containing a vertex have the required connected residues. At an edge, the residue consists of its two endpoints, its two seed vertices, and its two seed faces, with every cross-type pair incident. It is the join of three two-point sets; flags containing an edge therefore also have connected residues whenever their corank is at least two.

The residue at a single higher object (v,S) contains the connected Cayley graph on the affine space S with directions $\{q_e : e \in B_v\}$. Every remaining higher object in this residue contains a common vector with S by the definition of incidence, and hence attaches to this connected lower truncation. The same holds for a type-3 object. For an incident pair (v,S),(f,T), its residue is precisely the vertex-edge graph on the affine intersection $S \cap T$ with directions $\{q_e : e \in B_v \cap D_f\}$. Hypothesis 3 makes it connected. Finally, the global lower graph is connected because all q_e span N, and every higher object contains lower vertices. These cases exhaust all flags of corank at least two in rank four.

Regularity. Translation by N and the induced K-action on N and on the seed labels preserve incidence. The chamber parametrization above shows that $N \rtimes K$ acts freely and transitively: translate the vector, then use the regular action of K on seed chambers. A thin, residually connected geometry of finite rank is chamber connected (induct on rank inside residues, using connected incidence graphs). An automorphism fixing a chamber fixes each adjacent chamber by thinness and hence every chamber. Therefore any type-preserving automorphism is the unique element of G with its prescribed action on a base chamber.

The two leaf conditions. Hypothesis 1 makes the lower graph simple, which is (B1). An edge with endpoints in $S = n + W_B$ has direction $q_e \in W_B$; hypothesis 2 says exactly that $e \in B$, and then it is incident with the higher object by definition. This is (B2).

3 The seed and the binary quotient

Use permutations of {1,…,7}:

$$c = (1\,2)(3\,4)(6\,7), \quad b = (1\,5)(2\,3)(6\,7), \quad d = (1\,3)(2\,4)(6\,7).$$

Let $K = \langle c, b, d\rangle$. The exact checker obtains $|K| = 120$, $|cb| = |bd| = 5$, $cd = dc$, and all 64 subgroup-IP meets. The permutation group is $A_5 \times C_2$, although that name is not needed in the proof. Take the seed vertex, edge and face objects to be the left cosets of $\langle b, d\rangle$, $\langle c, d\rangle$ and $\langle c, b\rangle$, with incidence defined by nonempty intersection. The checker enumerates this incidence system itself: 12 vertices, 30 edges, 12 faces, 355 flags and 120 chambers; every flag extends to a chamber, every panel has size two, every residue of corank at least two is connected, and K is regular on chambers. Thus this is an actual regular map of type {5,5}.

Each vertex-star and each face-boundary contains five edges. There are 60 incident vertex-face pairs, each with exactly two common edges; nonincident pairs have no common edge. Define C to be the span of the characteristic vectors of these 24 five-edge blocks in $\mathbb{F}_2^{30}$. This definition is K-invariant because K permutes the two block families; the checker also verifies invariance under c,b,d explicitly.

Exact binary row reduction gives $\dim C = 17$, $\dim N = 13$, and the following complete small certificate:

- 30 nonzero, pairwise distinct quotient columns;
- 24 block spans of dimension four;
- all 720 column-in-block-span membership tests satisfy hypothesis 2;
- all 60 incident block-span intersections equal their two-column spans, of dimension two.

The concrete permutation marking, coset labels, code basis, quotient columns and block lists are included in the checker output. These are finite mathematical data. Neither the code computation nor its certificate substitutes for the global geometry proof in the affine construction theorem above.

Each five-column block has sum zero and rank four. Its only nonempty linear dependence is therefore the sum of all five columns. In particular its affine Cayley graph has a simple five-cycle and is nonbipartite. At an incident higher pair, the common shadow has two independent directions and hence is a square.

4 Distinguished involutions and sizes

Choose the seed base chamber at the identity cosets, and let $e_0 = \langle c, d\rangle$. Let a be translation by q_{e_0}, and regard b,c,d as elements of the complement K in $G = N \rtimes K$. The chamber-adjacent involutions of Γ, in type order, are (a,b,c,d). Indeed a exchanges the endpoints of its affine edge, b exchanges the two seed edges at the prescribed vertex-face pair, c changes the seed vertex, and d changes the seed face.

The edge stabilizer $\langle c, d\rangle$ fixes q_{e_0}, so a commutes with c,d. The two columns q_{e_0},q_{be_0} are distinct, and $(ab)^2$ is their nonzero sum, a translation of order two. Thus

$$|ab| = 4, \quad |bc| = |bd| = 5, \quad |ac| = |ad| = |cd| = 2.$$

The diagram is a star with centre b and labels 4,5,5. In particular a is a nondegenerate leaf of the required kind. Conjugates of a by K give all column translations, so these four involutions generate G. This marking has actual subgroup IP because it is the distinguished marking of the regular hypertope just proved, not because an arbitrary involutory marking is assumed to define a geometry.

The object and chamber counts are

$$|N| = 8192, \qquad |G| = 983040,$$

$$\text{object counts} = (8192, 122880, 6144, 6144), \qquad \text{chambers} = 983040.$$

The standard maximal parabolics have orders 120,8,160,160. For example a higher-object stabilizer is its 16-element translation space extended by its 10-element seed stabilizer. The base edge stabilizer has order eight. An incident pair of higher objects has a square residue, with eight chambers; hence there are 983040/8=122880 such pairs.

5 The geometric failure after halving

The global lower graph is nonbipartite, since each higher shadow already contains a five-cycle. All individual higher shadows are nonbipartite, so Construction 3.6 of the inspected source does not split any higher object by parity. Its two new lower types are the two labelled copies of N, with incidence $(n, 0) \sim (m, 1)$ precisely when n,m were adjacent in the original lower graph. Incidence with each unsplit higher object is affine containment in either copy. Higher-higher incidence is unchanged here.

Fix any incident pair of higher objects. Its original common shadow is a square. Its new rank-two residue is the canonical bipartite double cover of that square, namely two disjoint squares. Thus the new incidence geometry is not residually connected. There are exactly 122880 such failed residues. This is a literal local description, not an inference from a numerical graph test. The original affine construction has already established geometry, thinness, residual connectivity, regularity, (B1), (B2), and the leaf diagram independently.

For clarity, the transformed incidence system is still a geometry and is thin: a flag of higher objects has a common original vector, and, for a higher pair, a common adjacent original pair; each transformed flag extends using the original chamber description. The diagonal action of G on both labelled lower copies is regular on transformed chambers, again indexed by oriented original edges and their two seed higher objects. The rank-two residues above have more than one chamber-connected component; a free and transitive action on the global chambers does not repair those disconnected residues.

6 A direct subgroup-IP failure of the halved marking

Put $a' = aba$ and $H = \langle a', b, c, d\rangle$. In the local group $U_c = \langle a, b, c\rangle$, the subgroup $\langle b, c\rangle$ is dihedral of order ten and is transitive on the five edges of the base face. The translation $a'b$ is $q_{e_0} + q_{be_0}$. Consider the orbit of the unordered pair $\{e_0, be_0\}$ under this transitive degree-five action. Its graph is connected: all components have the same size dividing five, and the existence of an edge rules out singleton components. Pair differences along a connected graph span all even sums of its coordinate vectors. Thus the conjugates of $a'b$ span the even sums of the five face columns. Since the sum of all five columns is zero, these even sums already span the whole four-dimensional space. Consequently $a \in \langle a', b, c\rangle$ and

$$\langle a', b, c\rangle = \langle a, b, c\rangle, \qquad |\langle a, b, c\rangle| = 160.$$

The same argument with $\langle b, d\rangle$ and the base vertex-star gives

$$\langle a', b, d\rangle = \langle a, b, d\rangle, \qquad |\langle a, b, d\rangle| = 160.$$

In particular $H = G$. Actual IP for the original regular marking gives

$$\langle a', b, c\rangle \cap \langle a', b, d\rangle = \langle a, b\rangle, \qquad |\langle a, b\rangle| = 8.$$

But a' and b are distinct commuting involutions, since $(ab)^4 = 1$, and $\langle a', b\rangle$ has order four. It does not contain a: in the dihedral group $\langle a, b\rangle$ of order eight, the subgroup $\langle aba, b\rangle$ consists of the elements having an even number of occurrences of a. Thus a is an explicit witness to the failed meet. The final checker also constructs both local semidirect groups and this strict intersection directly.

This obstruction is precisely the nonprincipal recovery family from the general halving criterion: a is recovered after adjoining c or d separately, but not before either is adjoined. Here the original incidence system is an actual regular hypertope, unlike the earlier S_6 pregeometry separator.

7 Source and dependency boundary of the regular example

The inspected primary is Piedade–Tranchida, arXiv:2405.19050v1 (29 May 2024). Its exact input identity is recorded with this assembly. The source's broad Theorem A and its residual-connectivity assertion in Proposition 3.13 / Theorem 3.18 fail on this example, now even with regular thin input. In §4, the sentence immediately before Theorem 4.4 says the resulting object must be a regular hypertope; this claimed consequence also fails, as does the assertion that the halved distinguished generators must form a C-group.

The literal sentences of Theorem B and Theorem 4.4 assert a coset-isomorphism, a narrower statement. This note does **not** claim to refute that isomorphism. Indeed equality of the local new and old higher parabolics is compatible with the transformed geometry still having its asserted coset description while failing residual connectivity. We do not conflate those assertions.

The revised journal article, Combinatorica 45 (2025), article 10, DOI 10.1007/s00493-024-00134-y, has a later revision date. Its full revised body was not available in the bounded source audit. No claim about that body, absence of a subsequent correction, or novelty is made here.

The mathematical construction uses elementary permutation-group closure, binary linear algebra, the explicitly proved affine construction theorem, and the elementary chamber-connectivity property of finite-rank thin residually connected geometries. It uses no subgroup catalogue, no presumed quotient-IP theorem, and no full enumeration of the 983040-element group or the global incidence system. The retained parity terminology and source identification are separate from the direct square-residue and local-subgroup proofs above.

8 The complete finite certificate

The complete program `phase30_affine_map_check.py` produces the saved certificate. It uses Python's standard library only. The whole program and its saved exact output accompany this paper. The

replay command is recorded in the assembly guide.

The completed check enumerates all 355 seed flags, their extensions and residues, all 64 seed subgroup meets, all 720 column memberships, all 60 literal four-element span intersections, 72 code-generator and 51 code-basis invariance tests, and the two old and two changed local semidirect groups of order 160. It verifies all six marked product orders and the explicit excess-intersection witness a. It also constructs the eight-vertex doubled square and its two four-vertex components. The JSON includes both 30-coordinate quotient representatives and compressed 13-bit columns, all block labels, a basis of the code, and the induced generator action on the quotient.

The independent parent checker `phase30_regular_halving_root_check.py` uses $A_5 \times C_2$ product coordinates instead of seven-point permutations and separately checks the same seed geometry, binary hypotheses and local subgroup failure. Neither script enumerates the whole 983040-element group or its global incidence system. Global geometric conclusions and their derived counts depend on the affine construction and local recovery proofs above. The earlier exploratory probe is excluded from the final authored export manifest. The translation lattices of cubic toroids give a direct way to compute their halving groups. One first takes the subgroup generated by the new reflections in the infinite affine group, and then intersects its translation lattice with the original period lattice. This gives all the groups in Propositions 5.8–5.13 of the inspected preprint [1] by one calculation. It also detects missing relations in several of the claimed presentations. We distinguish the calculation of a marked group from the geometric conditions required to perform a halving.

9 Affine notation and the two subgroups

Let $n \geq 3$. Let B_n be the signed permutation group on $\mathbb{Z}^n$, and let B_n^+ be its subgroup with an even number of negative signs. Thus $|B_n| = 2^n n!$ and $|B_n^+| = 2^{n-1} n!$. Let s_i exchange coordinates $i, i+1$, let z_i change the sign of coordinate i, and let τ_v denote translation by v. We multiply affine transformations by composition, with the rightmost factor acting first. The usual cubic marking of $W = \mathbb{Z}^n \rtimes B_n$ is

$$a = \tau_{e_1} z_1, \qquad s_1, \dots, s_{n-1}, \qquad z_n.$$

We retain its standard affine Coxeter presentation $[4, 3^{n-2}, 4]$. This is the same classical realization used in Theorem 5.1 of [1]. Put

$$D = \{v \in \mathbb{Z}^n : \textstyle\sum_i v_i \equiv 0 \pmod 2\}, \quad x = a s_1 a = \tau_{e_1+e_2} z_1 z_2 s_1, \quad y = z_n s_{n-1} z_n = z_{n-1} z_n s_{n-1}.$$

The new markings after one and two endpoint operations generate

$$K = \langle x, s_1, \dots, s_{n-1}, z_n \rangle, \qquad J = \langle x, s_1, \dots, s_{n-1}, y \rangle.$$

Lemma 1. *One has $K = D \rtimes B_n$ and $J = D \rtimes B_n^+$. The Coxeter presentations of these groups are, respectively, the diagram obtained by halving the first endpoint of $[4, 3^{n-2}, 4]$, and the diagram obtained by halving both endpoints. In rank four the latter is a square of bonds labelled three; in rank five it is the four-leaf diagram with all bonds labelled three.*

Proof. The linear generators of K generate B_n. Multiplying x by $z_1 z_2 s_1$ gives $\tau_{e_1+e_2}$. Its conjugates by B_n are translations by all vectors $\pm e_i \pm e_j$, which span D. Conversely every displayed generator belongs to $D \rtimes B_n$. For J, the generators s_i and y generate B_n^+: the adjacent swaps give all permutations, while $y s_{n-1}$ changes two signs and its conjugates give every pair of sign changes. Again $z_1 z_2 s_1 \in B_n^+$, so $\tau_{e_1+e_2} \in J$. The orbit of this vector under B_n^+ contains all $\pm e_i \pm e_j$ when $n \geq 3$. To change just one of its two signs one changes the sign of a third, zero coordinate as well. This proves the second equality.

Here is also a presentation argument, needed when comparing the printed extra relations. In a Coxeter presentation with an endpoint a joined to b by a bond four and commuting with the remaining generators, replace a by $x = aba$. The proposed new Coxeter diagram duplicates the bonds of b at x, and makes x, b commute. Its abstract Coxeter group has an involutory automorphism exchanging x, b and fixing the other

generators. Adjoin an involution a acting by this automorphism. Eliminate x using $x = aba$. The relation $(xb)^2 = 1$ becomes $(ab)^4 = 1$; the remaining relations at x are conjugates of those at b. All other relations are exactly the old Coxeter relations. Thus the old group is the semidirect product of the new Coxeter group by $\langle a \rangle$, and the new group embeds as the kernel of the endpoint parity character. Apply this argument first at a in W, then at z_n in K. In the second application the character is the parity of the number of negative signs. This gives precisely the groups already identified and also covers $n = 3, 4$. □

10 The period lattice is intersected, not merely squared

For $s \geq 2$, the three toroid period lattices are

$$\Lambda_1 = s\mathbb{Z}^n, \qquad \Lambda_2 = sD, \qquad \Lambda_n = s(2\mathbb{Z}^n + \mathbb{Z}\mathbf{1}), \qquad \mathbf{1} = (1, \dots, 1).$$

These are the spans of the signed permutation orbits of se_1, $s(e_1 + e_2)$ and $s\mathbf{1}$, respectively, as in [1]. They are normal translation subgroups of W. Let $\overline{K}$ and $\overline{J}$ denote the images of the two marked groups in W/Λ_k.

Theorem 2. *Set $L = D \cap \Lambda_k$. Then, with their displayed affine markings,*

$$\overline{K} = (D/L) \rtimes B_n, \qquad \overline{J} = (D/L) \rtimes B_n^+.$$

The exact lattice and its index in D are as follows.

parameters	L	$[D : L]$
$k = 1$, *s even*	$s\mathbb{Z}^n$	$s^n/2$
$k = 1$, *s odd*	sD	s^n
$k = 2$	sD	s^n
$k = n$, *ns even*	$s(2\mathbb{Z}^n + \mathbb{Z}\mathbf{1})$	$2^{n-2}s^n$
$k = n$, *ns odd*	$2s\mathbb{Z}^n$	$2^{n-1}s^n$

In particular $|\overline{K}| = 2^n n![D : L]$ and $|\overline{J}| = 2^{n-1} n![D : L]$.

Proof. Restriction of the quotient homomorphism $W \to W/\Lambda_k$ to either group in Lemma 1 has kernel $D \cap \Lambda_k$. The linear subgroup intersects this kernel trivially. It therefore remains a complement, proving the two semidirect product statements.

A vector sv has even coordinate sum automatically when s is even, and exactly when $v \in D$ when s is odd. This gives the first two rows. The third is immediate. Write an element of Λ_n as $s(2v + t\mathbf{1})$. Its coordinate sum modulo two is nst. If ns is even there is no restriction; otherwise t must be even, giving $2s\mathbb{Z}^n$. Finally $[\mathbb{Z}^n : D] = 2$ and $[\mathbb{Z}^n : 2\mathbb{Z}^n + \mathbb{Z}\mathbf{1}] = 2^{n-1}$. Multiplication of a full lattice by s multiplies its index by s^n. These facts give all the displayed indices. □

This is a complete group description, rather than only an order calculation. It also supplies finite presentations uniformly. Choose the basis of D

$$d_1 = e_1 + e_2, \quad d_2 = e_2 - e_1, \quad d_i = e_i - e_{i-1} \quad (3 \leq i \leq n).$$

Adjoin commuting translation generators $T_1, \dots, T_n$ to a standard presentation of the finite group $P = B_n$ or B_n^+. For each generator p of P impose

$$pT_j p^{-1} = \prod_{i=1}^{n} T_i^{c_{ij}(p)}, \qquad pd_j = \sum_i c_{ij}(p) d_i.$$

For a basis of L impose $\prod_i T_i^{\ell_i} = 1$ whenever $\sum_i \ell_i d_i$ is a member of that basis. Such a basis is se_i in the first row, sd_i in the second and third, $2se_1, \dots, 2se_{n-1}, s\mathbf{1}$ in the fourth, and $2se_i$ in the fifth. All these vectors lie in D under the stated parity assumptions, so their coefficients are integers. The conjugation relations move every word to the form $T^v p$. The multiplication law and the evident affine realization then show that these presentations give exactly $(D/L) \rtimes P$. The marked element x is $T_1 z_1 z_2 s_1$, and the other marked elements are the specified linear transformations. Thus the presentation retains the actual marking as well as the abstract group.

11 Comparison with the single extra relations

The defect in a proposed extra relation can now be tested inside the infinite groups, before taking any finite quotient. A pure translation τ_v normally generates the translations whose vectors lie in the integer span of the orbit of v under the linear complement. Translations conjugate one another trivially, so there are no additional conjugates to supply missing periods.

For clarity we evaluate every family of extra words used in Propositions 5.8–5.13. Put

$$u = as_1 \cdots s_{n-1} z_n, \qquad u(X_1, \dots, X_n) = (X_n + 1, X_1, \dots, X_{n-1}).$$

Thus $u^n = \tau_{\mathbf{1}}$. The words for $k = n$, both before and after the second halving, are u^2 written in the new generators. This follows by moving a past s_i for $i \geq 2$ in the first case; in the second, move z_n past s_i for $i < n - 1$ and replace $z_n s_{n-1} z_n$ by y. The resulting word is

$$x s_2 \cdots s_{n-1} s_1 s_2 \cdots s_{n-2} y.$$

Its power $ns/2$, when integral, is $\tau_{s\mathbf{1}}$; its power ns in the odd case is $\tau_{2s\mathbf{1}}$.

For $k = 1$, the single-halving word is

$$q = x s_2 \cdots s_{n-1} z_n s_{n-1} \cdots s_2 s_1 = \tau_{e_1+e_2} z_2.$$

The corresponding word after two halvings is

$$q' = x s_2 \cdots s_{n-1} y s_{n-2} \cdots s_2 s_1 = \tau_{e_1+e_2} z_2 z_n.$$

Both square to τ_{2e_1}. Thus their displayed powers are τ_{se_1} for even s and τ_{2se_1} for odd s.

For $k = 2$, put

$$A = s_2 \cdots s_{n-1} z_n s_{n-1} \cdots s_2 = z_2, \qquad A' = s_2 \cdots s_{n-1} y s_{n-2} \cdots s_2 = z_2 z_n.$$

These concern the first and second operation, respectively. The words xAs_1A and $xA's_1A'$ are both $\tau_{e_1+e_2}$. The rank-four word printed separately in Proposition 5.12 is $xs_2s_1ys_1s_2$; conjugating y by s_2s_1 shows its linear part is $z_1z_2s_1$, so it has the same value. Consequently the extra relator is the translation $\tau_{s(e_1+e_2)}$ in every $k = 2$ case.

Proposition 3. *Interpret the listed extra relations in [1] as complete presentations, as their proofs do. They have the following exact discrepancies. The factor is the order of the presented group divided by the order of the intended marked group.*

parameters	*one halving*	*two halvings*
$k = 1$, *s odd*	2^{n-1}	2^{n-1}
$k = n$, *ns odd*	2^{n-1}	2^{n-1}
$k = n$, *n even*	1	2
all other cases	1	1

The table compares presentations of marked groups. Geometric validity of the second operation is a separate hypothesis.

Proof. The signed permutation orbit of e_1 spans $\mathbb{Z}^n$ under either complement, and that of $e_1 + e_2$ spans D. The orbit of $\mathbf{1}$ under B_n spans $E = 2\mathbb{Z}^n + \mathbb{Z}\mathbf{1}$: changing one sign gives $2e_i$, and these vectors together with $\mathbf{1}$ also generate every orbit vector. Its orbit under B_n^+ instead spans

$$M = 2D + \mathbb{Z}\mathbf{1}.$$

Indeed differences from $\mathbf{1}$ are twice the characteristic vectors of even subsets; the two-element subsets already span D. If n is odd, then $\mathbf{1} - e_i \in D$, so $2e_i = 2\mathbf{1} - 2(\mathbf{1} - e_i) \in M$ and $M = E$. If n is even, M has index two in E. To see this, every generator $2e_i$ of $2\mathbb{Z}^n$ has the same coset modulo $2D$, so the index is

at most two. It is not one: an equality $2e_1 = 2d + t\mathbf{1}$, with $d \in D$, forces $t = 2h$ and then $e_1 = d + h\mathbf{1}$, contradicting parity of coordinate sums.

Let R be the normal translation lattice generated by the printed extra relation. For odd s and $k = 1$, one has $R = 2s\mathbb{Z}^n$, whereas $L = sD$. Their relative index is $[sD : 2s\mathbb{Z}^n] = 2^{n-1}$. For odd ns and $k = n$, one has $R = 2sE$ for either operation, whereas $L = 2s\mathbb{Z}^n$; again the relative index is 2^{n-1}. For even n and $k = n$, the first operation gives $R = sE = L$, while the second gives $R = sM$, of index two in $L = sE$. The other rows give $R = L$ by the same orbit calculations. Lemma 1 identifies the underlying Coxeter group exactly, so its quotient by the displayed relation is $(D/R) \rtimes P$. The factor is therefore exactly $[L : R]$, not merely a lower bound from a finite quotient. □

For example, at $n = 3, k = 1, s = 3$, the intended first halving group has order $3^3 \cdot 48 = 1296$. The displayed presentation instead has order $[D : 6\mathbb{Z}^3] \cdot 48 = 108 \cdot 48 = 5184$. The accompanying small diagnostic constructs both affine groups directly, verifies every Coxeter relation and the displayed extra word, and obtains these orders. The all-parameter conclusions above follow from the lattice proofs, not from this finite check.

12 Attribution and the geometric interface

The cubic tessellation, the three period lattices and the halving operations are prior work. The parity characters, the second isomorphism theorem, and the standard Coxeter presentations are also established ingredients. Our contribution here is their common application to the exact period intersection, the complete presentations that it supplies, and the comparison that identifies the missing relations in the inspected edition. We make no claim of publication priority or about a later journal revision. The assertions that particular iterated constructions remain hypertopes, and the full list of their diagrams, are proved in the geometric sections that follow, with the nondegeneracy and residue arguments explicitly supplied. This note supplies the geometric arguments behind Section 5 and Figures 5–8 of the inspected preprint of Piedade–Tranchida [1]. The constructions are theirs. We use the previously reviewed component formulation of halving [5], including its precise residual-connectivity condition, and the separately reviewed affine group calculation [7]. In particular a presentation or a Coxeter diagram will not be used to infer residual connectivity. All assertions about the source refer to arXiv:2405.19050v1, not to an uninspected later journal edition.

13 The cubical model and the retained halving theorem

Let $n \geq 3$, $s \geq 2$, and $k \in \{1, 2, n\}$. In the usual cubical tessellation of $\mathbb{R}^n$ put

$$D = \{v \in \mathbb{Z}^n : \sum v_i \equiv 0 \pmod 2\}, \qquad L = D \cap \Lambda_k,$$

$$\Lambda_1 = s\mathbb{Z}^n, \quad \Lambda_2 = sD, \quad \Lambda_n = s(2\mathbb{Z}^n + \mathbb{Z}\mathbf{1}).$$

A cubical face is a product of integral singleton coordinates and unit intervals. Its dimension is its number of interval coordinates. The original toroid has objects the Λ_k-orbits of faces of dimensions $0, \dots, n$, with incidence induced by containment. We retain the classical assertion that this is the regular polytope of Theorem 5.1 of [1], attributed there to [2, Theorem 6D4]. Its usual marking, acting on column coordinates, is

$$a = \tau_{e_1} z_1, \quad s_1, \dots, s_{n-1}, \quad z_n,$$

where s_i swaps coordinates $i, i+1$ and z_i changes the sign of coordinate i. This is the standard affine realization, not a new classification of cubic toroids.

We recall exactly the halving result used here. A selected vertex–edge truncation is required to be a simple graph Q (B1). For every other object, incidence with an edge must be equivalent to incidence with

both endpoints (B2). For a flag F of other objects let Q_F be its common graph shadow. If Q is nonbipartite, the additional condition is

$$\text{all singleton shadows } Q_x \ (x \in F) \text{ nonbipartite} \implies Q_F \text{ nonbipartite.} \tag{PC}$$

In the component construction one takes a component of the bipartite double cover of Q, and lifts every other object by the components of the double cover of its shadow. The resulting incidence system is a geometry; it is residually connected when Q is bipartite, and, when Q is nonbipartite, exactly when PC holds. Thinness and chamber transitivity are preserved. These statements, including the residue isomorphisms used to compute the diagram, are the retained results of [5]. The regular coset identification and full type-preserving automorphism group are supplied by [6]. Neither result permits one to omit PC in a nonbipartite step.

Write H_{ab} for the geometry after a operations at the left end and b at the right end, with H_{00} the original toroid. The notation will only be used for the sequences proved to exist below. Choices of the two equivalent leaves at a fork are identified by a type permutation. A "second operation at the left" means the leaf $(0, 2)$ after the first operation $(0, 1)$.

14 Small translations and local cubical incidence

The following elementary separation facts keep the quotient arguments explicit. Put $R = \{\pm e_i \pm e_j : i \neq j\}$.

Lemma 4. *Every nonzero vector of Λ_k or L has a coordinate of absolute value at least s. No two vertices of a unit cube are therefore identified. Also:*

1. *$2e_i \in \Lambda_k$ or $2e_i \in L$ occurs exactly when $(k, s) = (1, 2)$.*
2. *If $r, r' \in R$ are distinct and $r - r' \in L$, then $(k, s) = (1, 2)$ or $(k, s) = (2, 2)$. Conversely both parameter pairs allow such a coincidence.*
3. *Suppose neither of these two parameter pairs occurs. If two even vertices of one unit cube differ by a root modulo L, they differ by that root literally. Their unique square is a face of the cube. If two even vertices at root distance have a common odd unit-step neighbor modulo L, that neighbor is one of the other two corners of this square, after a common lift.*

The corresponding statements hold in the dual cubical grid of cube centers, translated by $\frac{1}{2}\mathbf{1}$.

Proof. All lattices are contained in $s\mathbb{Z}^n$. For (1), it remains only to check $s = 2$. The vector e_i is neither in D nor in $2\mathbb{Z}^n + \mathbb{Z}\mathbf{1}$ for $n \geq 3$.

For (2), every coordinate of $r - r'$ has absolute value at most two. Thus $s = 2$. If $r - r'$ is even coordinatewise, the supports of r, r' are equal, and its nonzero support has size at most two. In $2D$ a single coordinate ± 2 is excluded; a pair $\pm 2, \pm 2$ is allowed. In $2(2\mathbb{Z}^n + \mathbb{Z}\mathbf{1})$ neither a one-coordinate nor a two-coordinate vector of this form occurs, since $n \geq 3$. Passing to L cannot create a coincidence. Examples are $e_1 + e_2$ and $-e_1 + e_2$ for $(1, 2)$, and $e_1 + e_2$ and $-e_1 - e_2$ for $(2, 2)$.

For (3), lift the cube and its two vertices p, q. In $q - p - r \in L$, every coordinate has absolute value at most two. For $s \geq 3$ it is zero. For $s = 2$, a nonzero even coordinate can occur only on the two coordinates supporting r, and the remaining allowed lattice $2(2\mathbb{Z}^n + \mathbb{Z}\mathbf{1})$ contains no such nonzero vector. Hence $q - p = r$. The square has exactly these two varying coordinates, so is unique and contained in the cube.

For the common-neighbor assertion, first lift the neighbor w and one endpoint p so that $w - p$ is a unit vector. The other incidence gives $q - p \equiv \alpha + \beta \pmod{L}$, where α, β are signed unit vectors. This sum is zero, a root, or $\pm 2e_i$. The zero case cannot equal a root modulo L. Two roots are equal by (2). Finally $\pm 2e_i - r$ cannot belong to $s\mathbb{Z}^n$ for any $s \geq 2$: at least one coordinate belonging to the other coordinate of r has value ± 1. Thus the sum is the same root, and the two possible intermediate vertices are exactly the other square corners. Translation of the entire argument gives the dual statement. □

We will also use the following form of local lifting. Fix a unit cube C, or a cube in the dual grid. Its vertices and subfaces inject into the quotient: a nonzero period cannot move a vertex or a subface of C to another vertex or subface of C, since the coordinate differences have absolute value at most one. An incident chain of face orbits has a common lift by first fixing its largest face and then lifting the smaller faces inside it. These lifts are unique. Incidences between subfaces of that largest face are consequently literal containments. When a vertex and a chain of faces form a flag, the vertex has its unique lift inside the smallest face. These observations apply to primal and dual cubes, and will be used together with Lemma 4 when a flag has two distinct vertex types.

Proposition 5. *The original left and right leaves satisfy B1 and B2 exactly when $(k, s) \neq (1, 2)$. After the first left operation the geometry H_{10} has the following literal model modulo L:*

0	*even integral vertices*
1	*odd integral vertices*
$i \geq 2$	*original i-cubes.*

Incidence between types 0 and 1 is unit-step adjacency; all incidences with cubes and between cubes are containment. This geometry is a regular hypertope.

Proof. Unit-step edges can become parallel in the quotient only if two distinct signed unit vectors differ by a period. Because periods are in $s\mathbb{Z}^n$, this is exactly the case $2e_i \in \Lambda_k$, settled by Lemma 4(1). To prove B2, lift a cube and two of its vertices. If their difference is a unit vector modulo Λ_k, the error has coordinates of absolute value at most two and can have a nonzero even coordinate only on that single axis. The only possible period is again $\pm 2e_i$ at $(k, s) = (1, 2)$. Otherwise the edge is a literal edge of the cube. The forward implication is containment. Self-duality proves the right-end assertion.

The parity of the coordinate sum colors the infinite grid. Its color-preserving period subgroup is exactly L. The component construction on the quotient by Λ_k is consequently the displayed colored-vertex construction modulo L. More explicitly, if $\Lambda_k \subset D$ the quotient grid is bipartite and no objects are doubled; if an odd period exists, there are two cosets of L in Λ_k, giving the two vertex lifts and the two compatible lifts of each cube. The three displayed lattices lie in D exactly when ks is even; an odd period otherwise supplies an odd closed walk in the quotient grid. Every cube shadow is an ordinary cube graph and is bipartite; higher flags are chains of cubes and their shadows are the smallest cubes. Thus PC holds, including in the globally nonbipartite case. The retained halving theorem gives thinness, residual connectivity and regularity. □

15 A second operation at one end

Proposition 6. *In H_{10} the leaves $(0, 2)$ and $(1, 2)$ satisfy B1 and B2 exactly when $(k, s) \neq (2, 2)$, assuming $(k, s) \neq (1, 2)$ already. Under this condition the second-left graph is nonbipartite, PC holds, and its halving H_{20} is a regular hypertope. Its objects modulo L are*

$0, 2$	*two separate copies of the even integral vertices*
1	*odd integral vertices*
$i \geq 3$	*original i-cubes.*

The two even-vertex types are incident when their underlying vertices differ by a root in R; even and odd types are incident at unit distance; incidences with cubes and between cubes are containment.

Proof. Each square has exactly two even corners. Its resulting edge has a root as difference vector. Conversely a root specifies a unique square. Lemma 4(2) therefore proves simplicity except at the stated small period. For B2 at a higher cube use part (3) of that lemma. At an odd vertex w, the even neighbors are the $2n$ points $w \pm e_i$. They are distinct by part (1). Two such neighbors give a square through w exactly when they are on different axes. Two opposite neighbors on the same axis cannot differ by a root modulo L, by the last argument in part (3). This proves B2.

At $(k, s) = (2, 2)$ the squares on coordinates 1,2 with respective corner sets $\{0, e_1, e_2, e_1 + e_2\}$ and $\{0, -e_1, -e_2, -e_1 - e_2\}$ are distinct, but have the same two even corners modulo $2D$. Hence B1 fails. B2 fails too: the odd vertex e_1 is adjacent to both these even corners but is not a corner of the second square modulo $2D$.

The root graph is connected because R spans D, and contains the triangle $0, e_1 + e_2, e_1 + e_3$. Its three vertices remain distinct. The graph shadow of an m-cube, $m \geq 3$, is its demicube: one parity class of its corners, adjacent at Hamming distance two. It is connected and contains a triangle. Indeed, any even difference can be removed two coordinates at a time, and the three corners with difference supports $\emptyset, \{1, 2\}, \{1, 3\}$ form a triangle (with a translated choice for the other parity). The shadow at an odd vertex is the cross-polytope graph on $\{w \pm e_i\}$, with edges between nonopposite pairs. It is connected and contains a triangle for $n \geq 3$.

A higher flag is either a chain of cubes, whose common shadow is the demicube of its smallest member, or such a chain together with an odd vertex w. In the latter case its even neighbors inside the smallest m-cube are the m corners obtained from w by a single allowed step; they form K_m, with $m \geq 3$. If there is no cube the shadow is the full cross-polytope graph. Local lifting justifies these descriptions in the quotient. All of them are nonbipartite, proving PC. Thus every higher object has just one lift in the new component construction, while the even vertices acquire two copies. This gives precisely the asserted model and proves its regular-hypertope conclusion. □

There is a useful consequence of this proof. Any flag of just the three lower types in H_{20} has a common lift whose vertices span an original face of dimension at most two. A single vertex spans a point; an even–odd pair spans an edge; a pair of even types spans their unique square. If the third, odd, type occurs, Lemma 4 puts it at one of the other corners of that square. The span is unique. If a cube is also prescribed, the span lies inside its unique lift. For H_{10} the corresponding lower flags span at most an edge. We call this the *small-face property* below.

16 Operations at the opposite end

Cube centers form the dual grid $\frac{1}{2}\mathbf{1} + \mathbb{Z}^n$. Color a center $c = \frac{1}{2}\mathbf{1} + v$ by the parity of $\sum v_i$. This is well-defined modulo L. For a primal d-face F, the centers of the n-cubes containing F form the vertices of a dual cube of dimension $n - d$. A facet between two such cubes is a dual edge. These are literal local statements, also after quotienting by L, by the local-lifting observation.

Proposition 7. *One may perform the first right operation after H_{10} for all $n \geq 3$, and after H_{20} for $n \geq 4$. The selected graph is bipartite, B1 and B2 hold, and the resulting geometries H_{11} and H_{21} are regular hypertopes. Their two upper types $n, n-1$ are respectively the two colors of original n-cube centers. These types are incident precisely when the cubes share a facet. Lower objects retain their earlier model, and are incident with an upper cube when they are contained in it. The cubes previously of dimension $n-1$ are removed.*

Proof. In both cases the original n-cubes and $(n-1)$-cubes are still present: for H_{20} this uses $n - 1 \geq 3$. Their graph is the dual unit-step grid modulo L. It is simple by Lemma 4(1), and bipartite by the displayed coloring. B2 for a lower original cube is the primal/dual version of the edge argument in Proposition 5. For a lower vertex, including either copy of an even vertex, it is the same argument for the dual n-cube around that vertex. No other object types occur. Every higher shadow is a connected bipartite cube graph: for a flag without an original cube, use its small-face span; when there is a chain of original cubes, use the largest one. Their codimension is at least two. The component construction therefore splits the center graph into its two colors, does not duplicate any other object, and has precisely the model asserted. Residual connectivity follows from the bipartite case of the retained theorem, without an additional PC assumption. □

Proposition 8. *A second right operation is possible in H_{11} when $n \geq 4$ and in H_{21} when $n \geq 5$, provided $(k, s) \neq (2, 2)$. Its graph is nonbipartite, B1 and B2 hold, and PC holds. It therefore gives H_{12} (dual to H_{21}) and H_{22}, respectively, as regular hypertopes. The two new upper types $n, n-2$ are separate copies of*

the even centers, incident at root distance; type $n-1$ consists of odd centers. All remaining objects keep their earlier meanings.

Proof. The objects of type $n-2$ are still original $(n-2)$-faces under the rank assumptions. They have four incident n-cubes, two with even centers. Thus the selected graph is the root graph on the even dual vertices. The proofs of simplicity and B2 are exactly Lemma 4 in the dual grid, applied separately to each actual higher object: a remaining original face has a dual cube as its shadow, a lower vertex has a dual n-cube, and an odd center has its cross-polytope neighbor graph. Different copies of a lower vertex have the same singleton shadow, which causes no problem for B2.

For completeness consider an arbitrary higher flag, not just its singletons. First suppose it contains no odd center. If it contains original faces, their largest member F determines the common dual cube, since all the lower vertices are in F. Its dimension is $n-\dim F \geq 3$, because type $n-2$ is being halved and the remaining face dimensions are at most $n-3$. If the flag contains only lower vertex types, their small-face span has dimension $d \leq 1$ in H_{11} and $d \leq 2$ in H_{21}. The common even-center shadow is the demicube of a dual cube of dimension $n-d \geq 3$. This also covers a singleton. An empty flag has the whole root graph, which is connected and nonbipartite.

Now suppose an odd center c is in the flag. Lift its primal n-cube. Every prescribed lower vertex and every prescribed primal face lifts inside this cube. The small-face property and local lifting make these lifts mutually compatible. Let F be their largest face, or their small-face span if there is no prescribed face. A neighboring even-center cube contains F exactly when its common facet with the cube at c contains F. There are $n-\dim F$ such facets, so the shadow is the clique $K_{n-\dim F}$. Its size is at least three by the same bounds. If there are no other objects, the shadow is the full cross-polytope graph. All these graphs are connected and nonbipartite. This proves PC for every flag. The component construction now gives exactly the stated duplicate even-center model. The retained theorem proves the conclusion. □

These arguments also identify the order of the operations. Where both are defined, doing the first right operation before the second left operation yields the same H_{21} incidence system. One can either apply Proposition 8 by duality to H_{11}, or compare the displayed models: both have two even-vertex copies, odd vertices, the remaining primal cubes, and the two center colors, with exactly the incidences already specified. The same comparison identifies the two paths to H_{22}. Thus an arrow in a diagram of constructions does not rely merely on equality of the group orders.

17 Exact diagrams and exhaustion of leaf operations

The regular-marking theorem in [6] replaces a selected leaf generator r by rqr, where q is its neighbor. Therefore the markings of the models just proved have the following uniform form:

$$(\ell_a, s_1, \dots, s_{n-1}, u_b),$$

where

j	ℓ_j	u_j
0	$\tau_{e_1} z_1$	z_n
1	$\tau_{e_1+e_2} z_1 z_2 (1\,2)$	$z_{n-1} z_n (n-1\,n)$
2	$\tau_{e_1+e_3} z_1 z_3 (1\,3)$	$z_{n-2} z_n (n-2\,n)$.

Here $(i\ j)$ denotes a coordinate swap. The second row is the first conjugation at each end; conjugating the next inward swap gives the third. This also checks all overlap cases in small rank. Since the geometries are thin and residually connected, these product orders are the actual rank-two residue gonalities.

We give indexed edge tables to make the small-rank diagrams unambiguous. Unlisted pairs have order two. Let

$$P = \{\{i, i+1\} : 1 \leq i \leq n-2\}.$$

Its edges all have label three. In the next table, an edge listed in the last column has label four; all other listed edges have label three. Repetitions in a union do not add another edge.

state	edges of label three	edges of label four
00	P	$01,\ (n-1)n$
10	$P \cup \{02\}$	$(n-1)n$
11, $n=3$	$P \cup \{02, 03, 13\}$	$\varnothing$
11, $n \geq 4$	$P \cup \{02, (n-2)n\}$	$\varnothing$
20, $n=3$	$P \cup \{01, 02\}$	$03, 23$
20, $n \geq 4$	$P \cup \{01, 02, 03\}$	$(n-1)n$
21, $n=4$	$P \cup \{01, 02, 03, 04, 24\}$	$\varnothing$
21, $n \geq 5$	$P \cup \{01, 02, 03, (n-2)n\}$	$\varnothing$
22, $n=5$	$P \cup \{01, 02, 03, 05, 25, 35, 45\}$	$\varnothing$
22, $n \geq 6$	$P \cup \{01, 02, 03, (n-3)n, (n-2)n, (n-1)n\}$	$\varnothing$

For example, 02 means the unordered pair $\{0, 2\}$. The states 21 and 22 only occur at the ranks in the table.

Here is a direct verification of the product orders, independent of any drawn diagram. Interior swaps on disjoint supports commute; consecutive ones have product order three. A signed swap on coordinates i, j has product order three with a swap sharing exactly one coordinate, and commutes with a disjoint swap. The two opposite signed swaps on the same pair commute. A sign change on one coordinate of a signed swap has product order four. The same statements apply to the affine left reflections after translating their common fixed intersection to the origin. In the few overlapping left–right pairs one can check the displayed maps directly: for $n = 3$, $\ell_1 u_1$ has order three and $\ell_2 z_3$ has order four; for $n = 4$, ℓ_2 and u_1 have order three; for $n = 5$, ℓ_2 and u_2 have order three. The linear parts already have these orders. Thus the orders cannot decrease after a translation quotient. Pairs declared commuting are commuting as affine maps, so the table holds for every allowed period.

Theorem 9. *Exclude $(k, s) = (1, 2)$. Up to duality and exchange of the two equivalent fork leaves, the complete sequence of admissible leaf halvings has states*

$$00 \longrightarrow 10 \longrightarrow 11,$$

and, if also $(k, s) \neq (2, 2)$,

$$10 \longrightarrow 20, \qquad 20 \longrightarrow 21 \longleftarrow 11 \quad (n \geq 4), \qquad 21 \longrightarrow 22 \quad (n \geq 5).$$

The initial arrow uses the bipartition construction if ks is even and the partition construction if ks is odd. The arrows at the first opposite end are bipartition operations. Each second operation at an end is a nonbipartite partition operation. For $n = 3$, states 20 and 11 are terminal; for $n = 4$, state 21 is terminal; for $n \geq 5$, state 22 is terminal. Every permitted state is a regular hypertope, with the exact diagram in the table.

Proof. Existence, nondegeneracy, parity and PC for all arrows have been proved in Propositions 5–8 and their dual forms. The original left endpoint reflection a exchanges vertex parity and the two left fork markings. At the other end z_n preserves vertex parity and exchanges center parity; it conjugates the two right fork markings. Whenever a second operation at the opposite end is already present, the stated rank bounds make these endpoint reflections commute with its altered marker. Thus these equivalences still apply at that stage. The explicit models identify both orders of independent operations.

It remains to justify exhaustion. The table shows that 00 has only its two endpoints as leaves. State 10 has its two left fork leaves and its right endpoint. At $(k, s) = (2, 2)$ the former fail B1 by Proposition 6; the latter still works. State 20 has no leaf when $n = 3$, and otherwise only the right endpoint. State 11 has no leaf when $n = 3$; otherwise its four fork leaves lead, up to the symmetries already described, to 21. State 21 has no leaf when $n = 4$, and otherwise just the two right fork leaves. Finally every vertex in state 22 has degree at least two. These are exactly the states and arrows stated. At the excluded second-step period, the same dual root coincidence prevents the fork operations in 11 as well. There is therefore no additional leaf operation omitted from the classification. □

This is the precise interpretation of the families in Figures 5–8: the second-step arrows require the extra small-period exclusion; the last two graphs are the overlapping-end cases in ranks four and five, not disjoint copies of the large-rank picture. The table specifies every bond even where a reproduced figure is difficult to read. The proof supplies the nondegeneracy and PC checks which the final paragraph of Section 5.3 leaves implicit. It does not restore the insufficient exact presentations in Propositions 5.8–5.13: the corrected marked groups and all missing period relations are treated separately in [7].

18 Scope and attribution

The toroids, the halving operation, the proposed diagram families, and their attribution to earlier polytope and diagram geometry are retained from the source. The new argument here is an explicit coordinate verification of every allowed geometric branch, using the previously reviewed parity-shadow criterion rather than the source's overbroad residual-connectivity claim. It is a Section 5 recovery component. It makes no whole-original admission, numerical compression claim, new conjecture claim or formalization claim.

19 Flag transitivity does not descend

Theorem A in the introduction of Piedade and Tranchida, *Constructing new geometries: a generalized approach to halving for hypertopes*, arXiv:2405.19050v1, asserts that the halved geometry is flag-transitive if and only if the original geometry is. We give a counterexample to the converse. The forward assertion in the detailed Proposition 3.14 is unaffected. This example is not thin; it is separate from the regular counterexample to residual connectivity.

Lemma 10. *Let R be a simple graph on V, and let g be an automorphism with $g^2 = 1$. Suppose x is not adjacent to gx for every $x \in V$. Define Q by $x \sim_Q y$ if and only if $x \sim_R gy$. Then Q is simple and undirected. The canonical bipartite double covers of Q and R are isomorphic by the identity on the first shore and by g on the second shore.*

Proof. Symmetry follows from $x \sim_R gy$ if and only if $gx \sim_R y$, and the hypothesis excludes a loop. In the double cover, x on the first shore is incident with y on the second precisely when $x \sim_R gy$. Thus the stated shore relabeling is the asserted type-preserving isomorphism. □

Take V to be the two-element subsets of $\{1, 2, 3, 4, 5\}$, let R have adjacency given by disjointness, and put $g = (12)$. Every $X \in V$ meets gX: if X contains exactly one of $1, 2$, its other element is fixed by g, and otherwise $X = gX$. Lemma 10 therefore applies. It is useful to write

$$a = \{1, 2\}, \quad p_i = \{1, i\}, \quad q_i = \{2, i\}, \quad b_i = \{3, 4, 5\} \setminus \{i\} \qquad (i = 3, 4, 5).$$

The complete edge list of Q is

$$\{p_i, p_j\},\ \{q_i, q_j\}\ (i < j), \qquad \{a, b_i\},\ \{b_i, p_i\},\ \{b_i, q_i\}\ (i = 3, 4, 5).$$

Indeed these are exactly the pairs X, Y for which $X \cap gY$ is empty, by separating the four displayed kinds of vertices. Thus Q is a connected cubic graph on ten vertices, with fifteen edges. The p_i and the q_i each form a triangle; a belongs to no triangle, since its three neighbors are the pairwise nonadjacent b_i. In particular Q is nonbipartite and not vertex-transitive.

Define a rank-three incidence geometry Γ as follows. Its objects of type 0 are the vertices of Q; its objects of type 1 are the edges of Q, with the usual endpoint incidence. Its objects of type 2 are two distinct elements z_+, z_-, each incident with every object of the other two types. Distinct objects of the same type are not incident.

Proposition 11. *The geometry Γ is residually connected and satisfies the source conditions B1 and B2. Its halving $\mathcal{P}(\Gamma)$ is residually connected and flag-transitive, whereas Γ is not flag-transitive. It has 60 chambers, as does its halving.*

Proof. Every flag extends to a chamber (v, e, z_ϵ) with $v \in e$. At z_ϵ the rank-two residue is the vertex-edge incidence graph of Q, hence is connected. At a vertex it is $K_{3,2}$; at an edge it is $K_{2,2}$. The whole incidence graph is connected because either type-2 object is incident with all lower objects. These are every residue of rank at least two, proving residual connectivity. Simplicity of Q is B1. Every type-2 object is incident with both endpoints of every edge and with that edge, so B2 holds. Flag transitivity would imply vertex transitivity of Q, which its triangle distinction excludes. Finally there are $15 \cdot 2 \cdot 2 = 60$ chambers.

We now identify the halving directly from its definition. Its types 0 and 1 are two copies of V, with X on the first shore incident with Y on the second when $X \cap gY = \emptyset$. The shadow of either z_ϵ is the whole connected nonbipartite graph Q. Its canonical double cover is connected, so z_ϵ has one lifted object, incident with every object of the first two types. Thus $\mathcal{P}(\Gamma)$ is the join of the bipartite double cover of Q with $\{z_+, z_-\}$.

For completeness, connectedness of that double cover follows without an enumeration. In a connected nonbipartite graph, a walk between two prescribed vertices can have either parity: traverse a path to an odd cycle, go around that cycle, and return along the same path to reverse the parity. Lifting these walks connects either choice of sheet over either vertex. The residues in the new geometry are therefore connected: at z_ϵ use this double cover, and at either lower object the residue is $K_{3,2}$. These again exhaust the rank-two residues.

After the second-shore relabeling $Y \mapsto gY$, the lower incidence relation is simply disjointness of two-element subsets. The group S_5 is transitive on ordered pairs of disjoint two-element sets: map the first pair, the second pair, and their unique remaining element separately. Consequently its action in the original labels is

$$X_0 \mapsto (\sigma X)_0, \qquad Y_1 \mapsto (g\sigma g Y)_1 \qquad (\sigma \in S_5).$$

It is transitive on the thirty incident lower pairs. Independently interchanging z_+ and z_- proves transitivity on all sixty chambers. Since every flag extends to a chamber, this is flag transitivity. The chamber count is $30 \cdot 2 = 60$. □

All nonempty higher shadows here equal Q and are nonbipartite, so the corrected parity condition is satisfied. The counterexample therefore does not exploit the separate failure of unconditional residual connectivity. Its only target is the reverse implication in the introductory flag-transitivity equivalence. The converse fails because an automorphism of the bipartite incidence geometry may act differently on its two shores, and need not descend to an automorphism of Q. The construction and proof use only elementary graph and incidence arguments; no novelty or publication-priority claim is made.

References

[1] C. A. Piedade and P. Tranchida, *Constructing new geometries: a generalized approach to halving for hypertopes*, arXiv:2405.19050v1 (29 May 2024). Only this full text is in the present source contract.

[2] P. McMullen and E. Schulte, *Abstract Regular Polytopes*, Cambridge University Press, 2002, Theorem 6D4 and the standard affine Coxeter realization of cubic tessellations.

[3] Human Compression Project, *Changes of chamber generators and binary translation groups*, accepted Topic C common paper, v2. The whole paper, including its binary-module sections, is retained.

[4] Human Compression Project, *Residues and stabilizers in geometric halving*, accompanying common supplement.

[5] Human Compression Project, *Geometric halving through graph covers and parity-compatible shadows*, phase27 accepted proof, 10 October 2026; its complete geometric proof is retained in the old common paper and the present residue supplement.

[6] Human Compression Project, *Stabilizers in geometric halving*, accepted proof reproduced in the accompanying common supplement.

[7] Human Compression Project, *Affine groups after one and two halvings*, accepted proof reproduced in the affine sections of this application.

[8] Human Compression Project, *An actual regular leaf hypertope whose halving loses residual connectivity*, accepted phase30 proof and finite certificate; all seven sections are reworked in this application.